

Agriculture: Apple Pest Management Guidelines

General Properties of Fungicides Used in Apples

Common name (trade name)	Chemical class	Activity	Mode of action (FRAC number) ¹	Resistance potential	Comments
Bordeaux	inorganic	contact	multi-site (M 01)	low	
captan	phthalimide	contact	multi-site (M 04)	low	highly toxic to honey bee larvae
copper	inorganic	contact	multi-site (M 01)	low	
cyprodinil (Vangard)	anilinopyrimidine	contact, systemic on some crops	single-site (9)	high ⁷	
fenarimol (Rubigan)	DMI ² - pyrimidine	systemic (local)	single-site (3)	high	
kresoxim-methyl (Sovran)	QoI ³	contact and systemic	single-site (11)	high ⁷	
mancozeb (Dithane)	carbamate EBDC ⁴	contact	multi-site (M 03)	low	
mefenoxam (Ridomil Gold)	phenylamide	contact and systemic	single-site (4)	high ⁷	
myclobutanil (Rally)	DMI ² -triazole	systemic (local)	single-site (3)	high	
pyrimethanil (Scala)	anilinopyrimidine	contact, systemic on some crops	single-site (9)	high ⁷	
sulfur	inorganic	contact	multi-site (M 02)	low	highly toxic to native strains of western predatory mite (<i>Galendromus occidentalis</i>) and to parasites
thiophanate- methyl (Topsin-M)	MBC ⁶	systemic (local)	single-site (1)	high ⁷	
trifloxystrobin (Flint Extra)	QoI ³	contact and systemic	single-site (11)	high ⁷	
triflumizole (Procure)	DMI ² -imidazole	systemic (local)	single-site (3)	high	
ziram	carbamate (DMDC) ⁵	contact	multi-site (M 03)	low	

1	Group numbers are assigned by the Fungicide Resistance Action Committee(http://www.frac.info/) (FRAC) according to different modes of action. Fungicides with different group numbers are suitable to alternate in a resistance management program (for more information see www.frac.info). In California, make no more than one application of a fungicide with a mode-of-action group number associated with high resistance risk before rotating to a fungicide with a different mode-of-action group number; for other fungicides, make no more than two consecutive applications before rotating to fungicide with a different mode-of-action group number.
2	DMI = demethylation (sterol) inhibitor
3	QoI = quinone outside inhibitor (strobilurin).
4	EBDC = ethylene bisdithiocarbamate
5	DMDC = dimethyl dithiocarbamate
6	MBC = methyl benzimidazole carbamate
7	Resistance has been found in California for certain fungicides with a single-site mode of action. To reduce the risk of resistance development, take the mode of action into account when choosing a fungicide. At the beginning of a treatment program, use a fungicide with a multi-site mode of action; for subsequent applications rotate or mix fungicides with different mode of action FRAC numbers. Use labeled rates (preferably the upper range) of the single-site fungicides, and limit the total number of applications per season.

Acknowledgment: Adaskaveg et al., 2025. [Fungicides, Bactericides, Biocontrols, and Natural Products for Deciduous Tree Fruit and Nut, Citrus, Strawberry, and Vine Crops in California\(PDF/PMG/fungicideefficacytiming.pdf\)](#)

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